2. Vinblastine is a standard chemotherapeutic drug used to treat cancer. Because it interferes with the assembly of microtubules, its effectiveness must be related to
(A) disruption of mitotic spindle formation.
(B) suppression of cyclin production.
(C) myosin denaturation and inhibition of cleavage furrow formation.
(D) inhibition of DNA synthesis.

3. One difference between cancer cells and normal cells is that cancer cells
(A) are unable to synthesize DNA.
(B) are arrested at the S phase of the cell cycle.
(C) continue to divide even when they are tightly packed together.
(D) cannot function properly because they are affected by density-dependent inhibition.

4. The decline of MPF activity at the end of mitosis is due to
(A) the destruction of the protein kinase Cdk.
(B) decreased synthesis of Cdk.
(C) the degradation of cyclin.
(D) the accumulation of cyclin.

5. In the cells of some organisms, mitosis occurs without cytokinesis. This will result in
(A) cells with more than one nucleus.
(B) cells that are unusually small.
(C) cells lacking nuclei.
(D) cell cycles lacking an S phase.

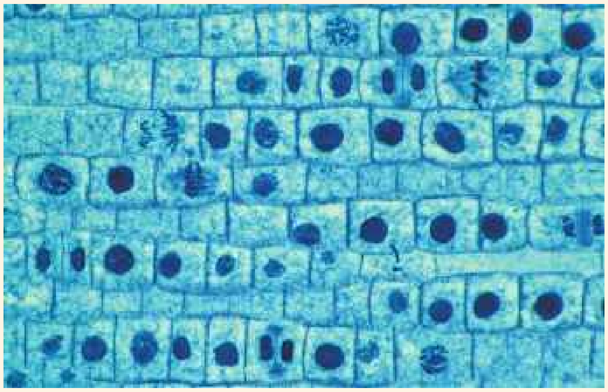
6. Which of the following occurs during S phase?
(A) condensation of the chromosomes
(B) replication of the DNA
(C) separation of sister chromatids
(D) spindle formation

Levels 3-4: Applying/Analyzing

7. Cell A has half as much DNA as cells B, C, and D in a mitotically active tissue. Cell A is most likely in
(A) G_1.
(B) G_2.
(C) prophase.
(D) metaphase.

8. The drug cytochalasin B blocks the function of actin. Which of the following aspects of the animal cell cycle would be most disrupted by cytochalasin B?
(A) spindle formation
(B) spindle attachment to kinetochores
(C) cell elongation during anaphase
(D) cleavage furrow formation and cytokinesis

9. **VISUAL SKILLS** The light micrograph shows dividing cells near the tip of an onion root. Identify a cell in each of the following stages: prophase, prometaphase, metaphase, anaphase, and telophase. Describe the major events occurring at each stage.



10. **DRAW IT** Draw one eukaryotic chromosome as it would appear during interphase, during each of the stages of mitosis, and during cytokinesis. Also draw and label the nuclear envelope and any microtubules attached to the chromosome(s).

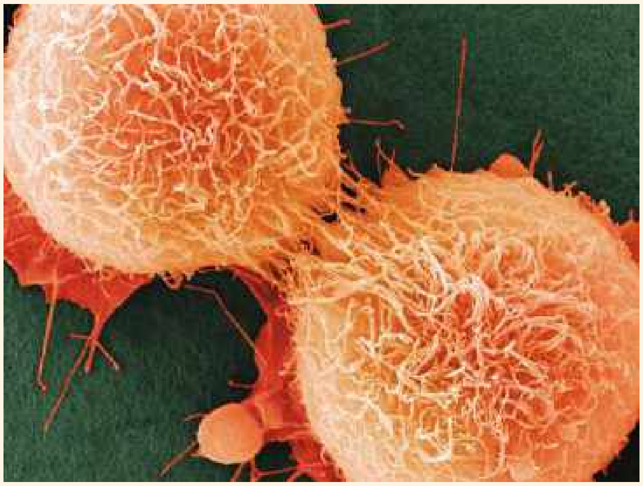
Levels 5-6: Evaluating/Creating

11. **EVOLUTION CONNECTION** The result of mitosis is that the daughter cells end up with the same number of chromosomes that the parent cell had. Another potential way to maintain the number of chromosomes would be to carry out cell division first and then duplicate the chromosomes in each daughter cell. Assess whether this would be an equally good way of organizing the cell cycle. Explain why evolution has not led to this alternative.

12. **SCIENTIFIC INQUIRY** Although both ends of a microtubule can gain or lose subunits, one end (called the plus end) polymerizes and depolymerizes at a higher rate than the other end (the minus end). For spindle microtubules, the plus ends are in the center of the spindle, and the minus ends are at the poles. Motor proteins that move along microtubules specialize in walking either toward the plus end or toward the minus end; the two types are called plus end–directed and minus end–directed motor proteins, respectively. Given what you know about chromosome movement and spindle changes during anaphase, predict which type of motor proteins would be present on (a) kinetochore microtubules and (b) nonkinetochore microtubules.

13. **WRITE ABOUT A THEME: INFORMATION** The continuity of life is based on heritable information in the form of DNA. In a short essay (100–150 words), explain how the process of mitosis faithfully parcels out exact copies of this heritable information in the production of genetically identical daughter cells.

14. **SYNTHESIZE YOUR KNOWLEDGE**



Shown here are two HeLa cancer cells that are just completing cytokinesis. Explain how the cell division of cancer cells like these is misregulated. Identify genetic and other changes that might have caused these cells to escape normal cell cycle regulation.

For selected answers, see Appendix A.